MIL-V-22862C

11 March 1976
SUPERSEDING
MIL-V-22862B
30 July 1968

MILITARY SPECIFICATION

# VALVES, COMPRESSOR SERVICE, REFRIGERANT-12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers manually operated compressor service valves for use with refrigerant-12.

1.2 Classification. Valves shall be of the following sizes as specified (see 6.2):

| Size | Tube OD (inches) | Bolt hole centers (inches) | Diameter of bolt holes (inches ± 1/64) | Minimum number of ports |
|---|---|---|---|---|
| 1 | 3/8 | 1-5/8 | 11/32 | 1 |
| 2 | 1/2 | 1-5/8 | 11/32 | 1 |
| 3 | 5/8 | 1-5/8 | 11/32 | 1 |
| 4 | 7/8 | 1-5/8 | 11/32 | 1 |
| 5 | 1-1/8 | 1-3/4 | 11/32 | 1 |
| 6 | 1-1/8 | 2-3/4 | 17/32 | 1 |

FSC 4820/4130

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## 2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-F-1421 - Flourocarbon Refrigerants.

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MS 17248 - Valves, Compressor Service, Refrigerant-12.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc. (ASHRAE)

Standard 15 - Safety Code for Mechanical Refrigeration

(Copies of this ASHRAE standard may be obtained from the American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc., 345 East 47th Street, New York, NY 10017.)



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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

* 3. REQUIREMENTS

3.1 Standard product. The valves delivered under this specification shall be manufacturer's commercial standard product, except for any changes necessary to comply with specification requirements.

3.2 Materials. Materials shall conform to the applicable specifications and requirements specified herein and in MS 17248. Materials not definitely specified shall be of the quality normally used for the purpose in commercial practice.

* 3.3 Design. The valves shall conform to MS 17248 and shall be designed for use with type 12 refrigerant conforming to BB-F-1421. The valves shall show no visible signs of leakage when subjected to a 350 psig refrigerant 12 pressure at 200 F (see 4.3.1).

3.4 Finish. The valves shall have the manufacturer's standard commercial finish.

3.5 Marking for identification. Valves shall be marked in accordance with the manufacturer's standard commercial practice.

* 3.6 Workmanship. The finished valves shall be clean and free of rust. All sharp edges, slivers, and burrs shall be removed. All scratches and tool marks on the exterior surface shall be removed.

* 4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The inspection lot shall consist of all valves of one size offered for inspection at one time. The sample unit shall be one completely fabricated valve.

* 4.2.2.1 Visual examination. The valves shall be examined for the defects in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 4.0, expressed in terms of defects per hundred units.

TABLE I. Classification of visual defects

| Examine | Defect |
|---|---|
| Standard product | Item not in accordance with manufacturer's standard product. |
| Finish | Dirt, oil, grease or rust.<br>Tool marks or scratches on the exterior. |
| Design | Not specified size. |
| Construction and workmanship | Surface rough, sharp edges, slivers or burrs. |
| Marking identification | Missing, illegible, incomplete or incorrect. |

4.2.2.2 Dimensional examination. Examination shall be made of the end item to determine compliance with dimensional requirements as specified in MS 17248. Any dimension not within the specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.



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4.2.2.3 Testing of the end item. The valves shall be tested as specified in 4.3.1 using inspection level S-3 with an AQL of 1.5, expressed in terms of defects per hundred units.

4.2.3 Inspection of preparation for delivery. Inspection of preparation for delivery shall be conducted in accordance with MIL-V-3.

4.3 Tests.

4.3.1 Leak tests. Valves shall be tested for leakage in accordance with SHRAE Standard 15 except that the test pressure shall be 350 psig at 00 F to determine compliance with 3.3. Noncompliance shall be cause for ejection of the valve and shall be recorded as a defect.

5. PACKAGING

5.1 Preservation - packaging. Preservation and packaging shall be level A r C as specified (see 6.2).

5.1.1 Levels A and C. Valves shall be preserved and packaged in accordance ith the applicable requirements of MIL-V-3.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. Valves, preserved and packaged as specified in .1, shall be packed in accordance with the applicable requirements of MIL-V-3.

5.3 Marking. In addition to any special marking required by the contract r order, interior packages and shipping containers shall be marked in accordance ith the applicable requirements of MIL-V-3.

6. NOTES

6.1 Intended use. Valves covered in this specification are intended rimarily for use on small size open- and hermetic-type compressors to permit solating the compressor for examination, charging, or obtaining pressure data.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of valve required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).



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6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 82

Preparing activity:

Army - GL

Project No. 4130-0142

Review activities:

Army - ME
GSA - GS



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DOCUMENT IDENTIFIER AND TITLE
MIL-V-22862C - VALVES, COMPRESSOR SERVICE, REFRIGERANT-12

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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